

Derivation of Eqs. 27 and 42-56 in Gauss, Stanton, and Bartlett, JCP 95, 2623 (1991)

$$\frac{\partial E}{\partial \chi} = \sum_{pq} D_{pq} \frac{\partial f_{pq}}{\partial \chi} + \sum_{pqrs} \Gamma_{pqrs} \frac{\partial \langle pq || rs \rangle}{\partial \chi}$$

Derivatives of f_{pq} and $\langle pq || rs \rangle$ contributions

$$\frac{\partial f_{pq}}{\partial \chi} = f_{pq}^{(x)} + U_{gp}^x f_{gg} + U_{pq}^x f_{pp} + \sum_t^{\text{all}} \sum_m^{\text{occ}} U_{tm}^x [\langle pt || gm \rangle + \langle pm || gt \rangle]$$

$$\frac{\partial \langle pq || rs \rangle}{\partial \chi} = \sum_t^{\text{all}} \left[U_{tp}^x \langle tg || rs \rangle + U_{tq}^x \langle pt || rs \rangle + U_{tr}^x \langle pq || ts \rangle + U_{ts}^x \langle pq || rt \rangle \right] + \langle pq || rs \rangle^{(x)}$$

$$\frac{\partial E}{\partial \chi} = \sum_{pq} D_{pq} \left[f_{pq}^{(x)} + U_{gp}^x f_{gg} + U_{pq}^x f_{pp} + \sum_t^{\text{all}} \sum_m^{\text{occ}} U_{tm}^x [\langle pt || gm \rangle + \langle pm || gt \rangle] \right] + \sum_{pqrs} \Gamma_{pqrs} \left[\sum_t^{\text{all}} \left\{ U_{tp}^x \langle tg || rs \rangle + U_{tq}^x \langle pt || rs \rangle + U_{tr}^x \langle pq || ts \rangle + U_{ts}^x \langle pq || rt \rangle \right\} + \langle pq || rs \rangle^{(x)} \right]$$

Collect terms with and without U^x contributions:

$$\frac{\partial E}{\partial \chi} = \sum_{pq} D_{pq} f_{pq}^{(x)} + \sum_{pqrs} \Gamma_{pqrs} \langle pq || rs \rangle^{(x)} + \sum_{pq} D_{pq} \left[U_{gp}^x f_{gg} + U_{pq}^x f_{pp} \right] + \sum_{pq} \sum_t^{\text{all}} \sum_m^{\text{occ}} D_{pq} U_{tm}^x [\langle pt || gm \rangle + \langle pm || gt \rangle] + \sum_{pqrs} \Gamma_{pqrs} \left[U_{tp}^x \langle tg || rs \rangle + U_{tq}^x \langle pt || rs \rangle + U_{tr}^x \langle pq || ts \rangle + U_{ts}^x \langle pq || rt \rangle \right]$$

Re-index U^x -dependent terms to use U_{pq}^x :

$$\frac{\partial E}{\partial \chi} = \sum_{pq} D_{pq} f_{pq}^{(x)} + \sum_{pqrs} \Gamma_{pqrs} \langle pq || rs \rangle^{(x)} + \sum_{pq} U_{pq}^x f_{pp} (D_{pq} + D_{qp}) + \sum_{p,q} U_{pq}^x \delta_{q,\text{occ}} \left[\sum_{rs} D_{rs} (\langle rp || sq \rangle + \langle rq || sp \rangle) + \sum_{rst} U_{pq}^x \left[\Gamma_{grst} \langle pr || st \rangle + \Gamma_{rgst} \langle rp || st \rangle + \Gamma_{rsqt} \langle rs || pt \rangle + \Gamma_{rstq} \langle rs || tp \rangle \right] \right]$$

Finally,
$$\frac{\partial E}{\partial \chi} = \sum_{pq} D_{pq} f_{pq}^{(x)} + \sum_{pqrs} \Gamma_{pqrs} \langle pq || rs \rangle^{(x)} - 2 \sum_{pq} U_{pq}^x I'_{pq}$$

where
$$I'_{pq} = -\frac{1}{2} \left[f_{pp} (D_{pq} + D_{qp}) + \sum_{rs} D_{rs} (\langle rp || sq \rangle + \langle rq || sp \rangle) \delta_{q,\text{occ}} + 4 \sum_{rst} \langle pr || st \rangle \Gamma_{grst} \right]$$

The expr. for I'_{pq} assumes Γ_{pqrs} is antisymmetric, but D_{pq} isn't necessarily symmetric. Also, the prefactor of $-\frac{1}{2}$ seems to be standard for most cc or ci gradient papers and leads to simpler forms. The above eqns. match Eqs. 42 and 43 from JCP 95, 2623 (1991).

Simplify orbital-gradient terms:

$$-2 \sum_{pq} U_{pq}^x I'_{pq} = -2 \sum_{ij}^{\text{occ}} U_{ij}^x I'_{ij} - 2 \sum_{ab}^{\text{vir}} U_{ab}^x I'_{ab} - 2 \sum_a^{\text{vir}} \sum_i^{\text{occ}} U_{ai}^x I'_{ai} - 2 \sum_i^{\text{occ}} \sum_a^{\text{vir}} U_{ia}^x I'_{ia}$$

Derivative of MO orthonormality conditions gives: $U_{pq}^x + U_{qp}^x + S_{pq}^x = 0$

This allows the last term above to be written as:

$$-2 \sum_i^{\text{occ}} \sum_a^{\text{vir}} U_{ia}^x I'_{ia} = +2 \sum_i^{\text{occ}} \sum_a^{\text{vir}} (U_{ai}^x + S_{ai}^x) I'_{ia}$$

The orbital-gradient term is thus:

$$-2 \sum_{pq} U_{pq}^x I'_{pq} = -2 \sum_{ij}^{\text{occ}} U_{ij}^x I'_{ij} - 2 \sum_{ab}^{\text{vir}} U_{ab}^x I'_{ab} + 2 \sum_a^{\text{vir}} \sum_i^{\text{occ}} U_{ai}^x (I'_{ia} - I'_{ai}) + 2 \sum_a^{\text{vir}} \sum_i^{\text{occ}} S_{ai}^x I'_{ia}$$

Using the non-canonical perturbed orbital conditions: $U_{ij}^x = -\frac{1}{2} S_{ij}^x$ and $U_{ab}^x = -\frac{1}{2} S_{ab}^x$

$$-2 \sum_{pq} U_{pq}^x I'_{pq} = \sum_{ij}^{\text{occ}} S_{ij}^x I'_{ij} + \sum_{ab}^{\text{vir}} S_{ab}^x I'_{ab} + 2 \sum_a^{\text{vir}} \sum_i^{\text{occ}} U_{ai}^x (I'_{ia} - I'_{ai}) + 2 \sum_a^{\text{vir}} \sum_i^{\text{occ}} S_{ai}^x I'_{ia}$$

All terms involving overlap derivatives may be collected together after splitting the last term:

$$2 \sum_a^{\text{vir}} \sum_i^{\text{occ}} S_{ai}^x I'_{ia} = \sum_a^{\text{vir}} \sum_i^{\text{occ}} S_{ai}^x I'_{ia} + \sum_a^{\text{vir}} \sum_i^{\text{occ}} S_{ia}^x I'_{ia}$$

$$\text{So, } \sum_{ij}^{\text{occ}} S_{ij}^x I'_{ij} + \sum_{ab}^{\text{vir}} S_{ab}^x I'_{ab} + \sum_i^{\text{occ}} \sum_a^{\text{vir}} S_{ia}^x I'_{ia} + \sum_a^{\text{vir}} \sum_i^{\text{occ}} S_{ai}^x I'_{ia} = \sum_{pq} S_{pq}^x I''_{pq}$$

$$\text{where } I''_{pq} = \begin{cases} I'_{qp} & \text{if } (p,q) = (a,i) \\ I'_{pq} & \text{otherwise} \end{cases}$$

The total gradient may therefore be written as:

$$\frac{\partial E}{\partial X} = \sum_{pq} D_{pq} f_{pq}(X) + \sum_{pqrs} \Gamma_{pqrs} \langle pq || rs \rangle^X + 2 \sum_a^{\text{vir}} \sum_i^{\text{occ}} U_{ai}^x X_{ai} + \sum_{pq} S_{pq}^x I''_{pq}$$

$$\text{where } X_{ai} = I'_{ia} - I'_{ai}$$

compare to Eq. 47
JCP 95, 2673 (1991)

The first-order CPHF equations for U_{ai}^x are:

$$-f_{ai}^{(x)} + S_{ai}^x \epsilon_i + \frac{1}{2} \sum_{j,k}^{occ} S_{jk}^x \left[\langle a_j || i k \rangle + \langle a_k || i j \rangle \right] = \sum_b \sum_j^{vir occ} U_{bj}^x \left[(\epsilon_a - \epsilon_i) \delta_{ab} \delta_{ij} + \langle a b || i j \rangle + \langle a j || i b \rangle \right]$$

Define $A_{ai,bj} \equiv (\epsilon_a - \epsilon_i) \delta_{ab} \delta_{ij} + \langle a b || i j \rangle + \langle a j || i b \rangle$

and $B_{ai}^x \equiv -f_{ai}^{(x)} + S_{ai}^x \epsilon_i + \frac{1}{2} \sum_{j,k}^{occ} S_{jk}^x \left[\langle a j || i k \rangle + \langle a k || i j \rangle \right]$

The CPHF eqs may be written more simply as:

$$\sum_b \sum_j^{vir occ} A_{ai,bj} U_{bj}^x = B_{ai}^x$$

or, in matrix-vector notation: $\underline{A} \underline{U}^x = \underline{B}^x$

The orbital contribution to the gradient is $\frac{\partial E}{\partial x} \leftarrow 2 \sum_a \sum_i^{vir occ} U_{ai}^x X_{ai}$

or, in matrix-vector notation: $\frac{\partial E}{\partial x} \leftarrow 2 \underline{X}^T \underline{U}^x$

Following Handy and Schaefer, JCP 81, 5031 (1984):

$$\underline{U}^x = \underline{A}^{-1} \underline{B}^x$$

$$\frac{\partial E}{\partial x} \leftarrow 2 \underline{X}^T \underline{A}^{-1} \underline{B}^x$$

cf. Eq. S1 of Gauss et al.

JCP 95, 2623 (1991)

same except for sign

Define

$$\underline{Z}^T \equiv \underline{X}^T \underline{A}^{-1} \rightarrow \underline{A}^T \underline{Z} = \underline{X} \text{ or } \sum_b \sum_j^{vir occ} A_{ai,bj}^T Z_{bj} = X_{ai}$$

$$\frac{\partial E}{\partial x} \leftarrow 2 \underline{Z}^T \underline{B}^x = 2 \sum_{ai} Z_{ai} B_{ai}^x$$

Expanding B_{ai}^x : $\frac{\partial E}{\partial x} \leftarrow -2 \sum_{ai} Z_{ai} f_{ai}^{(x)} + 2 \sum_{ai} Z_{ai} S_{ai}^x \epsilon_i + \sum_{ai} Z_{ai} \sum_{j,k}^{occ} S_{jk}^x \left[\langle a j || i k \rangle + \langle a k || i j \rangle \right]$

Reindexing the third term and splitting the first two:

$$\begin{aligned} \frac{\partial E}{\partial x} \leftarrow & - \sum_a \sum_i^{vir occ} Z_{ai} f_{ai}^{(x)} - \sum_a \sum_i^{vir occ} Z_{ai} f_{ia}^{(x)} + \sum_a \sum_i^{vir occ} Z_{ai} S_{ai}^x \epsilon_i + \sum_a \sum_i^{vir occ} Z_{ai} S_{ia}^x \epsilon_i \\ & + \sum_{ij}^{occ} S_{ij}^x \sum_a \sum_k^{vir occ} Z_{ak} \left[\langle a i || k j \rangle + \langle a j || k i \rangle \right] \end{aligned}$$

Finally, the total gradient expression becomes:

$$\frac{\partial E}{\partial \chi} = \sum_{p,q} D_{pq} f_{pq}^{(\chi)} + \sum_{p,q,r,s} \Gamma_{pqrs} \langle p q || r s \rangle^{\chi} + \sum_{p,q} I_{pq} S_{pq}^{\chi}$$

where $D_{ai} = D_{ia} = -Z_{ai}$

$$I_{ai} = I''_{ai} + Z_{ai} \epsilon_i$$

$$I_{ia} = I''_{ia} + Z_{ai} \epsilon_i$$

$$I_{ij} = I''_{ij} + \sum_{a' k} Z_{ak} [\langle a i || k j \rangle + \langle a j || k i \rangle]$$

$$I_{ab} = I''_{ab}$$

Note: To compare the above to Eqs. 52-56 of Gauss, Stanton, and Bartlett, JCP 95, 2623 (1991), be aware that the orbital z -vector equations I use differ by a sign from those in Eq. S1 of the JCP paper. Hence the signs on the z -vector contributions are reversed. Otherwise, everything matches exactly.